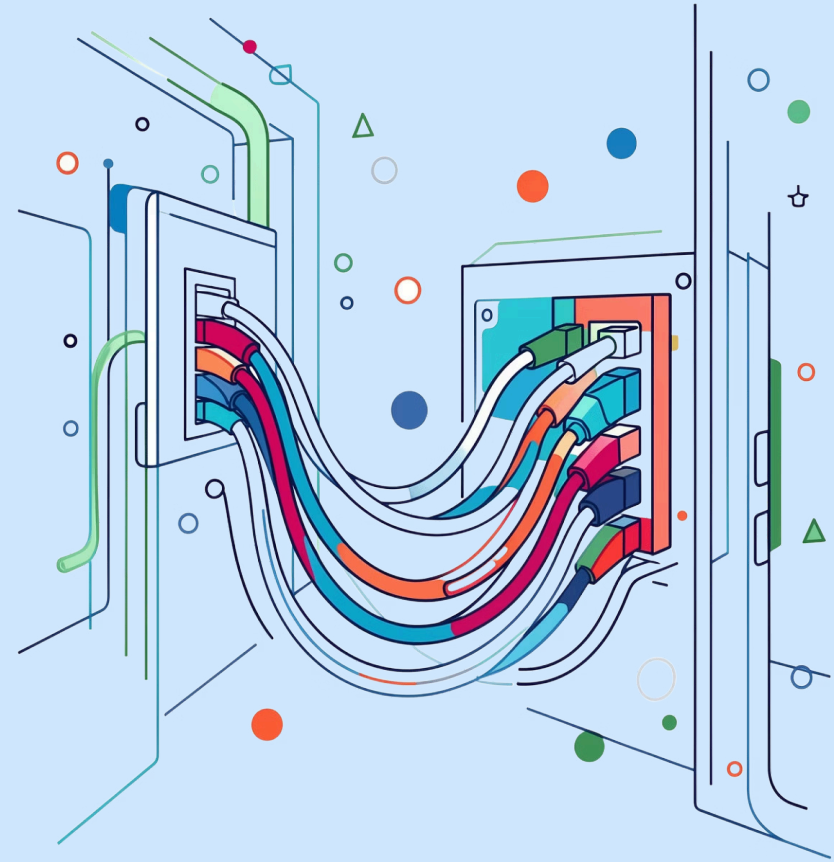


LAN Design & Data Flow Checkpoint

A hands-on walkthrough of MAC addressing, IP subnetting, frame structure, collision handling, routing, and NAT — using a realistic company-floor network scenario.

NETWORKING LAB CHECKPOINT



Your Scenario at a Glance

You are configuring a small network for a single company floor. Your design must support both wired and wireless devices sharing internet access through one router.



Wired Segment

- 1 Router (internet uplink)
- 2 Switches
- 4 Employee PCs
- 1 Network Printer



Wireless Segment

- 1 Wi-Fi Access Point
- 2 Wireless devices (laptops or phones)



What You'll Demonstrate

- MAC & frame handling
- IP addressing & subnetting
- Routing & packet forwarding

MAC Addresses: Assigning Layer 2 Identity

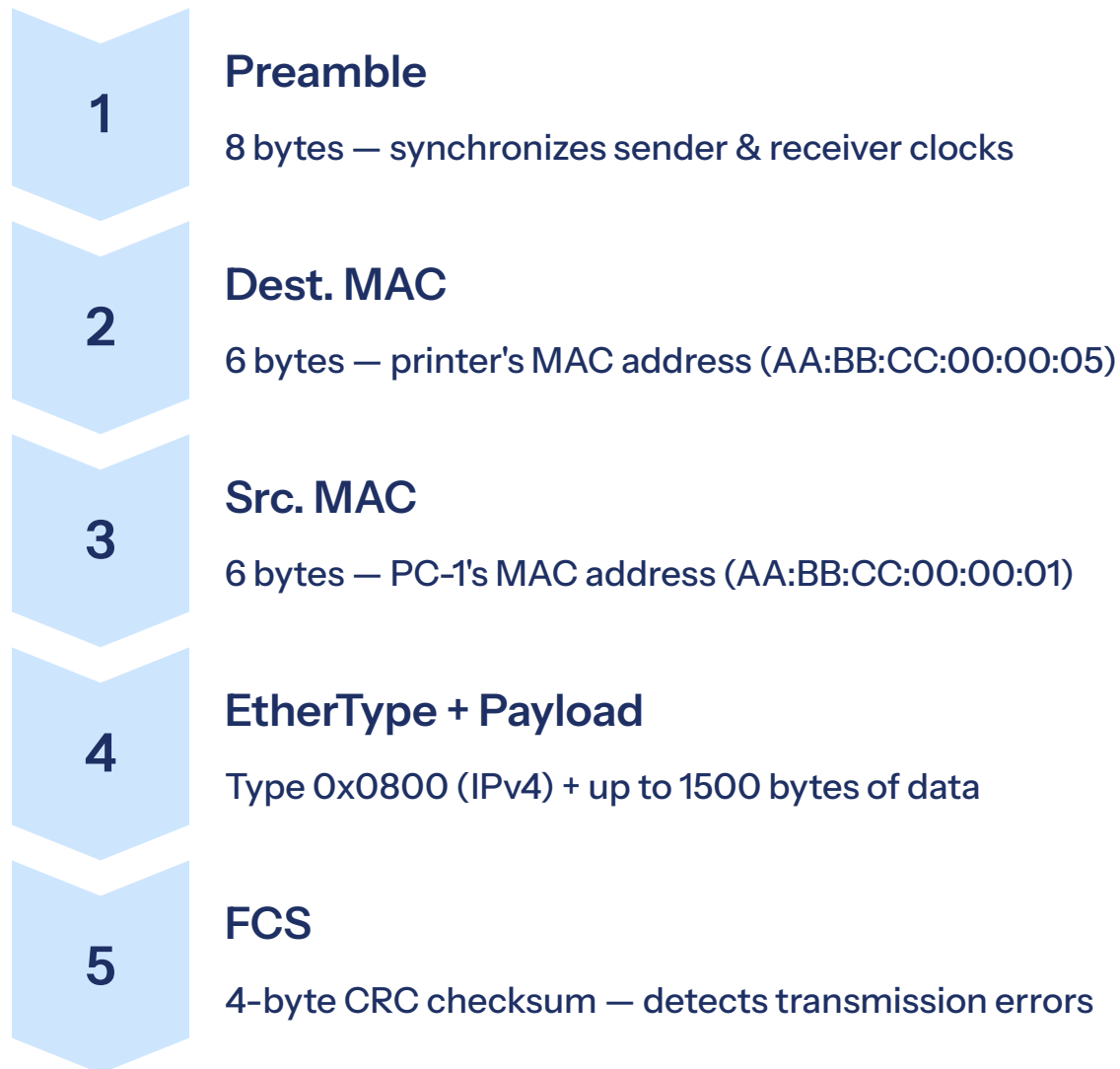
Every device on the network needs a unique **48-bit MAC address** (hardware address). Use the format `XX:XX:XX:XX:XX:XX`. You may invent addresses following the standard notation.

 Tip: Use a consistent prefix (e.g., `AA:BB:CC`) for your devices to keep your chart readable. The last three octets distinguish each device.

Device	Example MAC Address
PC-1	AA:BB:CC:00:00:01
PC-2	AA:BB:CC:00:00:02
PC-3	AA:BB:CC:00:00:03
PC-4	AA:BB:CC:00:00:04
Printer	AA:BB:CC:00:00:05
Wireless Device 1	AA:BB:CC:00:00:06
Wireless Device 2	AA:BB:CC:00:00:07
Router (LAN port)	AA:BB:CC:00:00:FF

Ethernet Frame Structure: PC to Printer

When PC-1 sends a print job to the printer, the data is wrapped in an **Ethernet II frame** at Layer 2. Each field has a specific role:



Collision Handling: Wired vs. Wireless

Ethernet (Wired) — CSMA/CD

Carrier Sense Multiple Access / Collision Detection

- Device listens before transmitting
- If a collision is detected, both devices stop
- Each waits a random backoff time before retrying
- Modern switched networks largely eliminate collisions

Wi-Fi (Wireless) — CSMA/CA

Carrier Sense Multiple Access / Collision Avoidance

- Devices cannot detect collisions while transmitting
- They *avoid* collisions by waiting a random backoff before sending
- Sender waits for an ACK; retransmits if none received
- RTS/CTS optional for hidden-node mitigation

- ❏ Key distinction: Wired Ethernet **detects** collisions after they occur; Wi-Fi **avoids** them beforehand since wireless transceivers cannot transmit and listen simultaneously.

IP Addressing & Subnetting

Using the private range 192.168.10.0/24, divide the network into two subnets — one for wired devices and one for wireless.

Subnet A — Wired (/25)

192.168.10.0 – 192.168.10.127

Device	IP Address
Router (gateway)	192.168.10.1
PC-1	192.168.10.10
PC-2	192.168.10.11
PC-3	192.168.10.12
PC-4	192.168.10.13
Printer	192.168.10.20

Subnet B — Wireless (/25)

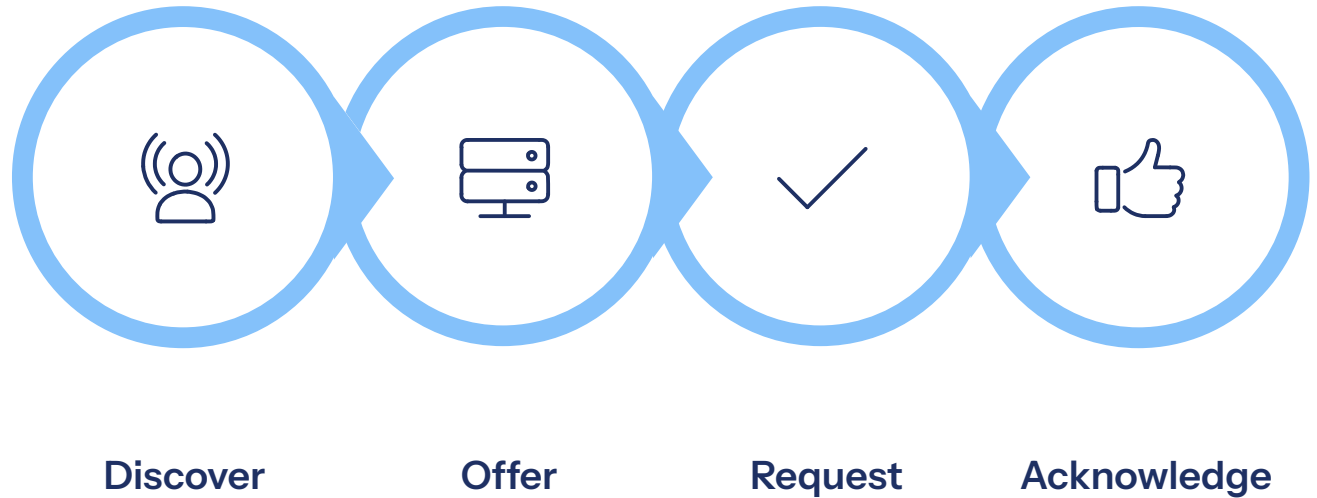
192.168.10.128 – 192.168.10.255

Device	IP Address
Access Point	192.168.10.129
Wireless Device 1	192.168.10.130
Wireless Device 2	192.168.10.131

 Default gateway for all devices: 192.168.10.1 (router LAN interface)

The Role of DHCP

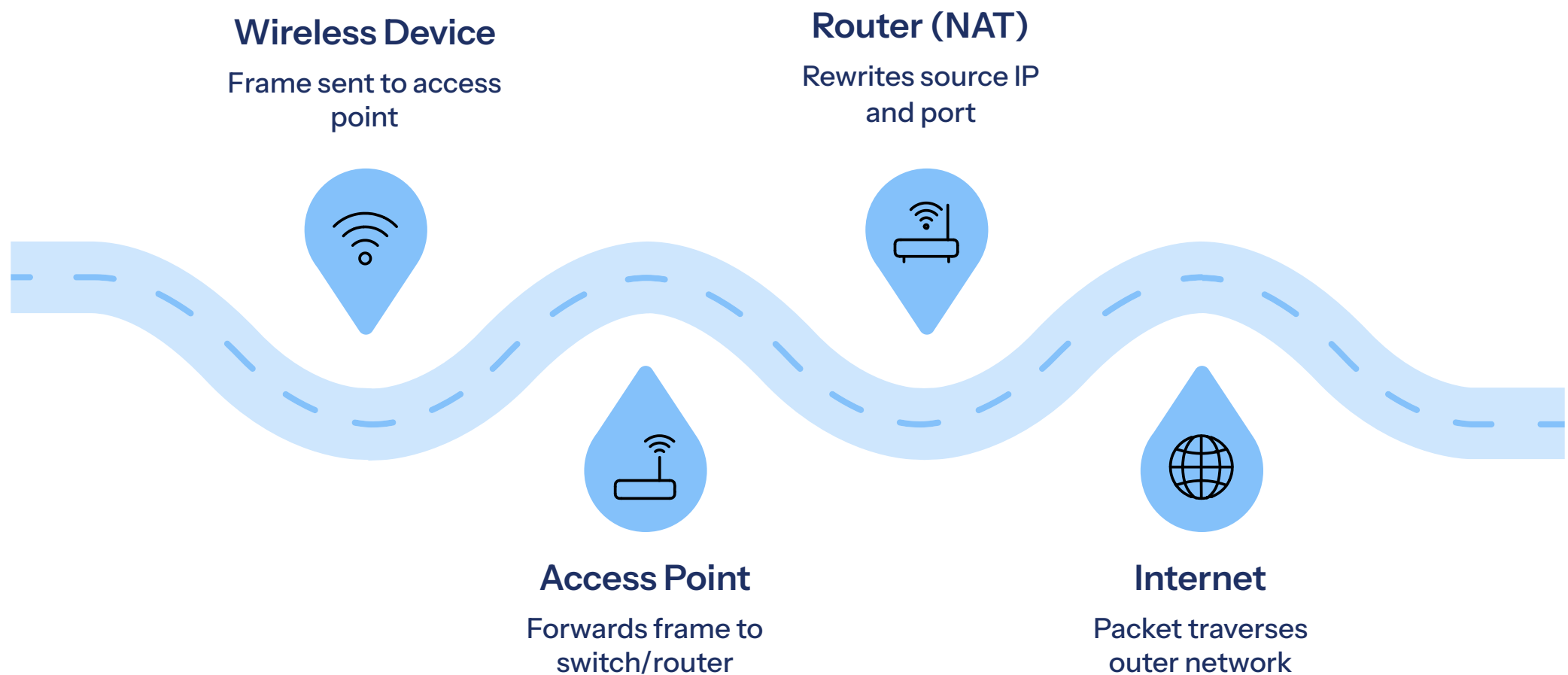
DHCP (Dynamic Host Configuration Protocol) automates IP address assignment so devices don't need manual configuration.



The router typically acts as the DHCP server, handing out IPs, the subnet mask, the default gateway, and DNS server addresses to each device when it joins the network.

Packet Path: Wireless Device to the Internet

When Wireless Device 1 opens a web page, its packet travels through several layers of the network before reaching the external server.



At each hop, the **IP header** provides the destination address so routers can make forwarding decisions, while **MAC addresses** are rewritten at every Layer 2 segment.

Routing, Next-Hop Decisions & NAT

How the Router Forwards Packets

Routing Table Lookup

Router checks the destination IP against its routing table using longest-prefix matching.

Next-Hop Selection

If no local match, the packet is forwarded to the ISP gateway (default route 0.0.0.0/0).

TTL Decrement

Each router decrements the TTL field by 1. If TTL reaches 0, the packet is dropped and an ICMP error is returned.

NAT: Network Address Translation

Because private IPs (192.168.10.x) are not routable on the internet, the router performs **PAT (Port Address Translation)**:

- Replaces the private source IP with the router's public IP
- Assigns a unique source port to track each session
- Maintains a NAT translation table for return traffic
- Inbound replies are de-translated back to the correct private host

✓ NAT conserves public IPv4 addresses and adds a layer of implicit security by hiding internal topology.

Checkpoint Deliverables Summary

01

Network Diagram

Show all devices, switches, router, AP, and connections with labeled MAC and IP addresses.

02

Frame Analysis

Describe the Ethernet frame from PC to printer with all fields populated.

03

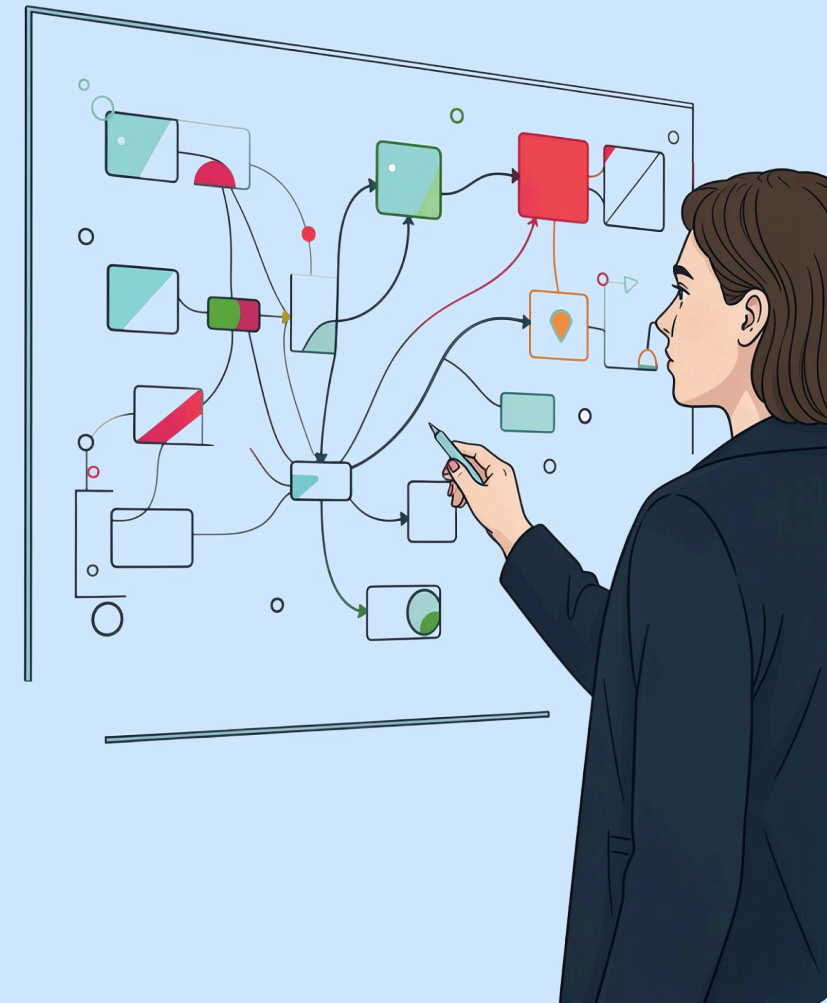
Subnet Table

Show both subnets with address ranges, assigned IPs, and default gateway.

04

Packet Walk-Through

Trace a wireless device's request to an external site, covering IP headers, routing decisions, and NAT translation.



- ❏ Submit as a Google Slides/PowerPoint presentation **or** a Google Doc/PDF report with clear diagrams and step-by-step explanations.